THE UNIVERSITY OF SYDNEY

**ELEC5211: Power System Dynamics and Control**

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# LAB REPORT 1

## GROUP 16

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# 目　　录

1. Introduction ..... 3
2. System Description and Parameters ..... 4
3. Mathematical Model Development ..... 5
3.1 Active and reactive power balance at the terminal bus ..... 5
3.2 Dynamic model of the SMIB system ..... 7
3.3 Power-Angle Relation ..... 9
4. Steady-State Operating Point Calculation ..... 10
4.1 Operating Conditions ..... 10
4.2 Initial Reactive Power and Current ..... 10
4.3 Internal Transient Voltage ..... 11
4.4 Initial Rotor Angle and Equilibrium Check ..... 11
5. Power-Angle Curve and Maximum Power Transfer ..... 13
5.1 Power-angle curve ..... 13
5.2 Maximum transferable power ..... 14
5.3 Stability interpretation ..... 14
6. Simulation Method ..... 16
6.1 Software and model ..... 16
6.2 State variables ..... 16
6.3 Disturbance setup ..... 16
6.4 Numerical solver ..... 17
7. Results and Discussion ..... 18
7.1 Response to step increase in mechanical power ..... 18
Case 1: Mechanical power exceeding $P_{max}$ ..... 18
Case 2: Step increase in mechanical power ..... 18
7.2 Discussion ..... 19
8. Conclusion ..... 20

# 1. Introduction

This experiment investigates the dynamic and stability characteristics of a Single Machine Infinite Bus (SMIB) system, which is the fundamental model for power system dynamic analysis. The core objectives of this experiment are:

(1) Derive the power balance equation of the generator terminal bus, and establish the dynamic model of the SMIB system.

(2) Analyze the power-angle characteristic of the generator, calculate the maximum transferable power, and verify the instability mechanism when the mechanical power exceeds the stability limit.

(3) Simulate the dynamic response of the system under a step increase in mechanical input power, and verify the small signal stability of the system.

(4) Understand the core principles of power system transient stability and small signal stability through numerical simulation, and master the application of digital simulation techniques in power system dynamic analysis.

(5) Clarify the assumptions and applicable scope of the classical constant E' model, and understand the role of excitation system in power system stability control.

# 2. System Description and Parameters

The system under study is a Single Machine Infinite Bus (SMIB) system, which consists of a synchronous generator connected to an infinite power grid through a step-up transformer and a transmission line. The infinite bus represents a power grid with infinite capacity, whose voltage and frequency are fixed regardless of the generator's output. The system parameters are listed in Table 1, all parameters are in per unit (p.u.) system.

| Parameter | Symbol | Value | Unit | Physical Meaning |
|---|---|---|---|---|
| Generator transient reactance | $X_G'$ | 0.25 | p.u. | Generator internal transient reactance |
| Generator inertia constant | H | 0.05 | s | Rotor inertia constant |
| Damping coefficient | D | 0.1 | p.u. | Rotor damping coefficient |
| Exciter gain | $K_v$ | 1 | p.u. | Excitation system gain |
| Transformer reactance | $X_{Th}$ | 0.25 | p.u. | Step-up transformer reactance |
| Transmission line reactance | $X_L$ | 0.5 | p.u. | Transmission line reactance |
| Infinite bus voltage | $V$ | 1.0 | p.u. | Infinite bus voltage magnitude |
| Excitation reference voltage | $V_{10}$ | 1.0 | p.u. | Excitation system reference voltage |
| System rated frequency | $f_0$ | 50 | Hz | System rated frequency |

# 3. Mathematical Model Development

This section develops the mathematical model of the Single Machine Infinite Bus (SMIB) system, which serves as the basis for the steady-state calculation, the power-angle analysis, and the time-domain simulation. The derivation is carried out in three steps. First, the active and reactive power balance equations at the generator terminal bus are established. Second, the rotor motion equation is introduced to describe the electromechanical dynamics of the generator. Third, the classical power-angle relation is obtained for the simplified model used in the subsequent analysis.

## 3.1 Active and reactive power balance at the terminal bus

To describe the power transfer in the SMIB system, the infinite-bus voltage is selected as the reference phasor. The infinite bus has fixed voltage magnitude and angle, while the generator terminal bus voltage varies according to the operating condition.

### 3.1.1 Reference Phasor Definition

The infinite bus is selected as the reference phasor because its voltage magnitude and angle remain constant during the analysis. Therefore, the infinite-bus voltage phasor is written as

$$V = V\angle 0$$

and the generator terminal-bus voltage phasor is written as

$$V_1 = V_1\angle\delta_1$$

where $V$ is the infinite-bus voltage magnitude, $V_1$ is the terminal-bus voltage magnitude, and $\delta_1$ is the phase angle of the terminal bus with respect to the infinite bus. The generator internal transient emf is written as

$$E' = E'\angle\delta$$

where $E'$ is the transient internal voltage magnitude and $\delta$ is the rotor angle.

### 3.1.2 Generator-Side Active and Reactive Power

The current flowing from the generator internal transient voltage source to the terminal bus is

$$I_G = \frac{E' - V_1}{jX_G'}$$

The complex power injected by the generator into the terminal bus is

$$S_G = P_G + jQ_G = V_1 I_G^*$$

Thus, the active and reactive powers supplied by the generator are

$$P_G = \frac{E'V_1}{X_G'} \sin(\delta - \delta_1) \quad (1)$$

$$Q_G = \frac{E'V_1}{X_G'} \cos(\delta - \delta_1) - \frac{V_1^2}{X_G'} \quad (2)$$

3.1.3 Network-Side Active and Reactive Power

The transformer and transmission line are represented by pure reactances. Their equivalent external reactance from the terminal bus to the infinite bus is

$$X_e = X_{Th} + X_L = 0.25 + 0.5 = 0.75\ p.u.$$

$X_{Th}$=transformer reactance

$X_L$=transmission line reactance

The current flowing from the terminal bus to the infinite bus is therefore

$$I_e = \frac{V_1 - V}{jX_e}$$

The complex power injected from the terminal bus into the external network is

$$S_{net} = P_{net} + jQ_{net} = V_1 I_e^*$$

Substituting the current expression into the complex power formula and separating the real and imaginary parts gives

$$P_{net} = \frac{V_1 V}{X_e} \sin \delta_1 \quad (3)$$

$$Q_{net} = \frac{V_1^2}{X_e} - \frac{V_1 V}{X_e} \cos \delta_1 \quad (4)$$

Equation (3) shows that the active power transfer is mainly determined by the voltage-angle difference, while Equation (4) shows that the reactive power transfer depends on both the voltage magnitudes and the angle difference.

3.1.4 Terminal-Bus Power Balance

Since no load is connected directly at the generator terminal bus, the power injected by the generator must be equal to the power delivered into the external network. Therefore, the terminal-bus power balance equations are

$$P_g = P_{net} \tag{5}$$

$$Q_g = Q_{net} \tag{6}$$

Substituting Equations (1)–(4) into Equations (5) and (6), the final active and reactive power balance equations can be written as

$$\frac{E'V_1}{X_G'}\sin(\delta - \delta_1) = \frac{V_1 V}{X_e}\sin\delta_1 \tag{7}$$

$$\frac{E'V_1}{X_G'}\cos(\delta - \delta_1) - \frac{V_1^2}{X_G'} = \frac{V_1^2}{X_e} - \frac{V_1 V}{X_e}\cos\delta_1 \tag{8}$$

These two equations describe the steady-state coupling between the generator internal voltage, terminal voltage, rotor angle, and the network variables.

## 3.2 Dynamic model of the SMIB system

The dynamic model describes how the system variables (power angle, rotor speed, internal potential) change over time when the system is disturbed. It consists of three core components: the rotor motion equation, the electromagnetic power equation, and the excitation system model.

3.2.1 Rotor Motion Equation (Core of Dynamic Analysis)

The rotor motion is governed by the swing equation. Let $\omega$ denote the rotor speed in per unit, $\omega_0$ the synchronous angular frequency, $P_m$ the mechanical input power, $P_e$ the electromagnetic output power, and D the damping coefficient.

| Symbol | Expression / Value | Unit | Meaning |
|---|---|---|---|
| $T_J$ | 2H=0.1 | s | Inertia time constant |
| $\omega_0$ | $2\pi f_0 = 314.1593$ | rad/s | Synchronous angular frequency |
| $\delta$ | — | rad | Rotor angle |
| $\omega$ | — | p.u. | Rotor speed |
| $P_m$ | — | p.u. | Mechanical input power |
| $P_e$ | — | p.u. | Electromagnetic output power |
| D | 0.1 | p.u. | Damping coefficient |

The inertia time constant is defined as

$$T_J = 2H \tag{9}$$

and the synchronous angular frequency is

$$\omega_0 = 2\pi f_0 \tag{10}$$

The swing equation can be written in first-order form as

$$\frac{d\delta}{dt} = \omega_0(\omega - 1) \tag{11}$$

$$T_J \frac{d\omega}{dt} = P_m - P_e - D(\omega - 1) \quad (12)$$

Equation (11) shows that the rotor angle increases when the rotor speed is higher than the synchronous speed. Equation (12) shows that the rotor accelerates when the mechanical input power exceeds the electrical output power, and decelerates otherwise.

For completeness, the same dynamics can also be expressed in second-order form as

$$\frac{T_J}{\omega_0}\frac{d^2\delta}{dt^2} = P_m - P_e - \frac{D}{\omega_0}\frac{d\delta}{dt} \quad (13)$$

3.2.2 Electromagnetic Power Equation

The electromagnetic power output of the generator is determined by the power angle characteristic, which is derived from the power balance equation for the total system reactance:

$$P_e = P_G = \frac{E'V_1}{X_G'}\sin(\delta - \delta_1) \quad (14)$$

This expression is coupled with the terminal-bus power balance equations and is suitable for the general SMIB model.

3.2.3 Excitation System Model

The excitation system regulates the generator's internal potential E' to maintain the terminal voltage at the reference value. This experiment uses a simplified fast excitation model:

$$\frac{dE'}{dt} = K_v(V_{10} - V_1) \quad (15)$$

However, in Tasks 3 and 4 of this experiment, the excitation system is assumed to maintain the transient internal voltage constant. Therefore,

$$\frac{dE'}{dt} = 0, \qquad E' = constant \quad (16)$$

Under this assumption, the model is reduced to a second-order classical SMIB model with two state variables, δand $\omega$.

## 3.3 Power-Angle Relation

For the simplified classical model, the generator transient internal voltage $E'$ is assumed constant, and the total reactance from the internal voltage to the infinite bus is

$$X_{\Sigma} = X_G{}' + X_{Th} + X_L \tag{17}$$

For the present system,

$$X_{\Sigma} = 0.25 + 0.25 + 0.5 = 1.0\ p.u. \tag{18}$$

The electromagnetic power can then be expressed in the standard power-angle form as

$$P_e(\delta) = \frac{E'V}{X_{\Sigma}} \sin\delta \tag{19}$$

Equation (19) is the classical power-angle relation of the SMIB system. It shows that the electrical output power depends sinusoidally on the rotor angle. The maximum transferable power is therefore

$$P_{max} = \frac{E'V}{X_{\Sigma}} \tag{20}$$

This power-angle relation forms the basis for the steady-state operating point calculation, the maximum power transfer analysis, and the interpretation of system stability in the following sections.

# 4. Steady-State Operating Point Calculation

To perform the power-angle analysis and the time-domain simulation, the initial steady-state operating point of the SMIB system must first be determined. According to the laboratory specification, the generator operates at its nominal active power output $P_0 = 1p.u.$, and the power factor at the infinite-bus side is $\cos\phi = 0.85$, with reactive power injected into the system. For Tasks 3 and 4, the transient internal voltage E' is assumed to remain constant, so the classical SMIB model is adopted. Under this assumption, the infinite-bus voltage magnitude is V = 1 p. u., and the total reactance between the generator internal transient voltage and the infinite bus is

$$X_\Sigma = X_G' + X_{Th} + X_L$$

Substituting the given parameters gives

$$X_\Sigma = 0.25 + 0.25 + 0.5 = 1.0\ p.u.$$

## 4.1 Operating Conditions

The initial active power output is specified as

$$P_0 = 1.0\ p.u.$$

and the power factor angle is

$$\phi = \cos^{-1}(0.85) = 31.79°$$

Since the generator injects reactive power into the system, the initial reactive power is positive and is calculated as

$$Q_0 = P_0 \tan\phi \tag{21}$$

which gives

$$Q_0 = 1.0 \times \tan(31.79°) = 0.6197\ p.u.$$

Therefore, the initial complex power at the infinite-bus side is

$$S_0 = P_0 + jQ_0 = 1 + j0.6197$$

## 4.2 Initial Reactive Power and Current

Using the complex power relation

$$S_0 = VI_0^*$$

and taking the infinite-bus voltage as the reference phasor $V = 1\angle 0$, the initial current is

$$I_0 = \left(\frac{S_0}{V}\right)^*$$

Since $V = 1\angle 0$, this becomes

$$I_0 = P_0 - jQ_0 \tag{22}$$

Hence,

$$I_0 = 1 - j0.6197$$

## 4.3 Internal Transient Voltage

For the classical SMIB model, the generator internal transient voltage is related to the infinite-bus voltage and the current by

$$E_0' = V + jX_\Sigma I_0 \tag{23}$$

Substituting V = 1, $X_\Sigma = 1.0$, and $I_0 = 1 - j0.6197$, we obtain

$$E_0' = 1 + j(1 - j0.6197)$$

which gives

$$E_0' = 1.6197 + j1.0$$

Therefore, the magnitude of the internal transient voltage is

$$E' = |E_0'|$$

and hence

$$E' = \sqrt{1.6197^2 + 1.0^2} = 1.9036 p.u.$$

## 4.4 Initial Rotor Angle and Equilibrium Check

The initial rotor angle is obtained from the phase angle of the internal transient voltage phasor:

$$\delta_0 = \angle E_0' \tag{24}$$

Thus,

$$\delta_0 = \tan^{-1}\left(\frac{1.0}{1.6197}\right) = 31.69°$$

At the steady-state operating point, the rotor speed is equal to the synchronous speed, so

$$\omega_0^{(p.u.)} = 1$$

and the mechanical input power must be equal to the electrical output power, that is,

$$P_{m0} = P_{e0} = P_0 = 1.0\ p.u. \tag{25}$$

Using the classical power-angle relation derived in Section 3.3,

$$P_e = \frac{E'V}{X_\Sigma}\sin\delta$$

the initial operating point can be verified as

$$P_{e0} = \frac{1.9036 \times 1}{1.0}\sin(31.69°) \approx 1.0\ p.u.$$

which confirms that the obtained operating point is consistent with the specified steady-state condition.

# 5. Power-Angle Curve and Maximum Power Transfer

Based on the steady-state operating point obtained in Section 4, the power-angle characteristic of the SMIB system can now be established. For Tasks 3 and 4, the transient internal voltage E′E'E′ is assumed to remain constant, and therefore the classical SMIB model is adopted. Under this assumption, the electrical output power depends on the rotor angle according to the power-angle relation derived in Section 3.3.

## 5.1 Power-angle curve

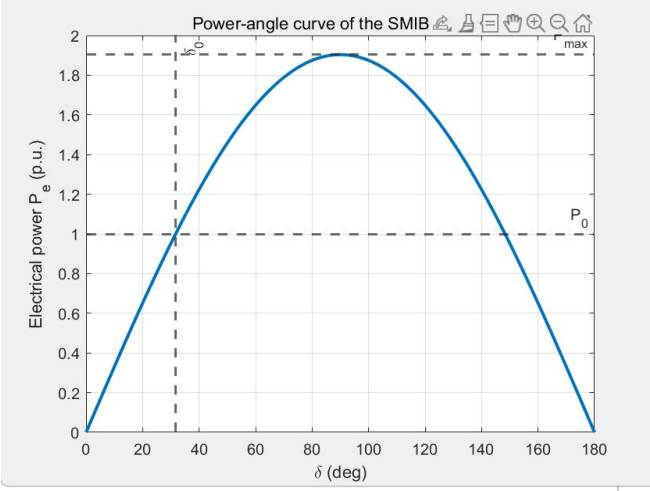



Figure 1. Power-angle curve of the SMIB system under the classical model with constant E'.

Using Equation(19), the electromagnetic power of the classical SMIB model is given by

$$P_e(\delta) = \frac{E'V}{X_\Sigma}\sin\delta \tag{26}$$

From Section 4, the steady-state calculation gives E' = 1.9036p. u. The infinite-bus voltage is V = 1.0p. u., and the total reactance is

$X_\Sigma = 1.0p.u.$ Therefore, Equation(26) becomes

$$P_e(\delta) = 1.9036\sin\delta \tag{27}$$

Equation (27) is the power-angle curve of the SMIB system. It shows that the electrical output power varies sinusoidally with the rotor angle. The initial operating

point obtained in Section 4 is

$$P_0 = 1.0\ p.u., \qquad \delta_0 = 31.69°$$

Since this operating point lies on the left-hand side of the peak of the power-angle curve, it corresponds to a feasible steady-state operating point.

## 5.2 Maximum transferable power

The maximum transferable power occurs when $\sin\delta = 1$, that is, when $\delta = 90°$. Therefore, the maximum power transfer capability of the SMIB system is

$$P_{max} = \frac{E'V}{X_\Sigma} \tag{28}$$

Substituting E' = 1.9036, V = 1.0 p. u., and $X_\Sigma = 1.0p.u.$ gives

$$P_{max} = 1.9036p.u.$$

This result indicates that the generator can transfer at most 1.9036 p.u. of electrical power under the given network condition. Since the initial mechanical and electrical power is only 1.01.01.0 p.u., the system operates below the maximum power limit and therefore has a valid equilibrium point.

## 5.3 Stability interpretation

The power-angle curve provides a direct physical interpretation of system stability. When the mechanical input power is lower than the maximum transferable power, that is,

$$P_m < P_{max} \tag{29}$$

the horizontal line representing the mechanical power intersects the power-angle curve, and a steady-state equilibrium point exists. In this experiment, the initial operating point satisfies this condition, so the system can remain in synchronism.

However, if the governor increases the mechanical input power beyond the peak of the power-angle curve, then

$$P_m > P_{max}$$

and no intersection between the mechanical power line and the electrical power curve exists. In that case, no steady-state equilibrium can be established. The accelerating power remains positive, the rotor continues to accelerate, and the rotor angle keeps increasing. As a result, the generator loses synchronism with the infinite bus.

Therefore, the power-angle curve not only determines the maximum transferable power, but also explains the instability mechanism when the mechanical input exceeds the transmission capability of the system.

# 6. Simulation Method

## 6.1 Software and model

The dynamic response of the SMIB system was simulated in MATLAB using the classical second-order model derived in Section 3. In this model, the transient internal voltage E' is assumed to remain constant, and the system is described by two state variables: the rotor angle $\delta$ and the rotor speed deviation $\Delta\omega$.

## 6.2 State variables

The SMIB system uses a second-order electromechanical dynamic model with two state variables:

(1) Rotor power angle δ: Phase difference between the generator transient EMF E′ and infinite bus voltage V, in radian or degree.

(2) Rotor speed deviation $\Delta\omega$: $\Delta\omega = \omega - \omega_0$, where $\omega_0 = 1$ p.u. is the synchronous speed, representing the generator rotor speed deviation.

These two variables fully describe the transient and steady-state electromechanical dynamics of the generator.

## 6.3 Disturbance setup

Two typical disturbance scenarios are configured to test system dynamic performance and stability:

(1) Step increase in mechanical power

At t= 1 s, the mechanical power input $P_m$ is increased from its steady-state value $P_{m0}$ to 110% $P_{m0}$ and remains constant. This tests the small-signal dynamic response and steady-state regulation performance.

(2) Over-power disturbance (beyond steady-state stability limit)

Mechanical power $P_m$ is set higher than the peak value of the power-angle curve $P_{emax}$. This is used to verify loss of synchronism and transient instability.

All disturbances are applied at steady-state initial conditions to ensure accurate dynamic response analysis.

## 6.4 Numerical solver

The system was solved numerically in MATLAB using ode45, which is suitable for nonlinear ordinary differential equations of the type considered in this experiment. The simulation interval was chosen to include both the transient process and the final post-disturbance behavior. In the implementation used for this report, the initial state was set as

$$x(0) = \begin{bmatrix} \delta_0 \\ 1 \end{bmatrix}$$

and the main outputs recorded from the simulation were:

(1) rotor angle $\delta$(t)

(2) speed deviation $\Delta\omega$(t)

(3) electrical power $P_e$(t)

(4) mechanical power $P_m(t)$

These quantities were used to assess whether the generator remained in synchronism after the disturbance and whether a new equilibrium point could be reached.

# 7. Results and Discussion

## 7.1 Response to step increase in mechanical power

### Case 1: Mechanical power exceeding $P_{max}$

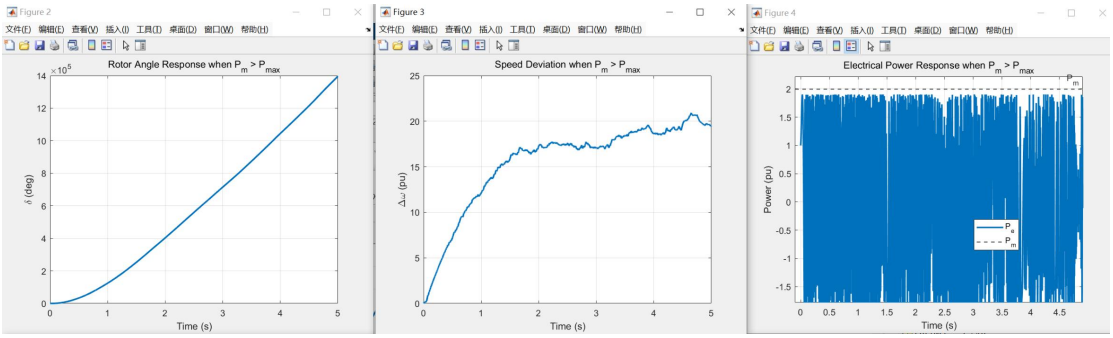



Figure 2. Dynamic responses when $P_m > P_{max}$

Figure 2 show the system response when the mechanical input power exceeds the peak of the power-angle curve(we use $P_m$=2 p.u.). The rotor angle $\delta$ increases continuously over time without reaching a steady value, indicating that the generator is accelerating and losing synchronism. The speed deviation $\Delta\omega$ does not return to zero and remains positive, which confirms that the rotor continues to accelerate. The electrical power $P_e$ oscillates due to the sinusoidal relationship with $\delta$, but it cannot reach the mechanical input power because it is limited by $P_{max}$. This demonstrates that no equilibrium point exists when $P_m > P_{max}$, and the system becomes unstable.

### Case 2: Step increase in mechanical power

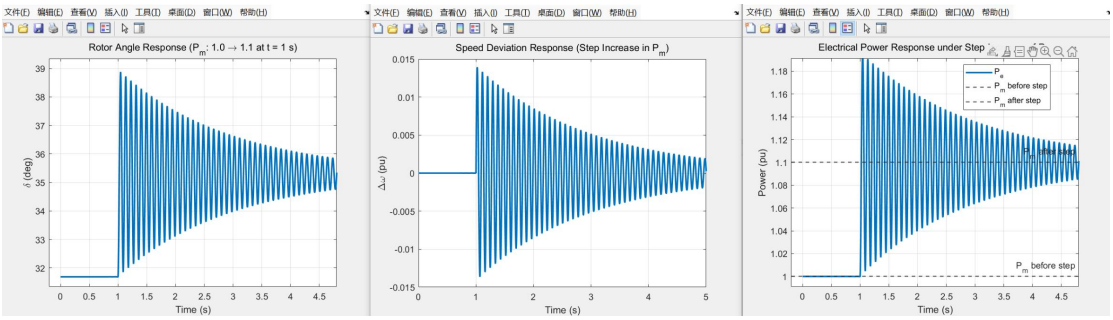



Figure 3. Dynamic responses under step increase in mechanical power

Figure 3 show the response when the mechanical power increases from 1.0 p.u.

to 1.1 p.u. at t = 1 s. Before the disturbance, the system operates at steady state with constant $\delta$ and zero $\Delta\omega$. After t = 1 s, the increase in Pm causes an imbalance between $P_m$ and $P_e$, resulting in acceleration. The rotor angle $\delta$ increases and exhibits oscillatory behavior. The speed deviation $\Delta\omega$ also shows oscillations. Due to damping, the oscillations gradually decrease, and $\Delta\omega$ the system settles to a new equilibrium. The electrical power $P_e$ increases and eventually matches the new mechanical input power.

This indicates that the system remains stable when $P_m < P_{max}$.

## 7.2 Discussion

The comparison between the two cases clearly shows the importance of the power-angle curve.

The maximum transferable power $P_{max}$ represents the stability limit of the system. When Pm exceeds this limit, no equilibrium exists and the system loses synchronism. When $P_m$ is below this limit, the system can reach a new stable operating point after disturbance.

The simulation results are consistent with the theoretical analysis.

# 8. Conclusion

This laboratory experiment successfully completed the dynamic analysis and control research of the Single Machine Infinite Bus (SMIB) system, and fully achieved the preset experimental objectives. Through theoretical derivation, modeling, numerical simulation and result analysis, we have deeply understood the steady-state and dynamic characteristics of the typical power system.

First, the active and reactive power balance equations at the generator terminal bus were rigorously derived based on Kirchhoff's voltage law and complex power conservation, which laid a theoretical foundation for subsequent system modeling and power calculation. On this basis, the complete dynamic model of the SMIB system was established, including the classical generator swing equation, constant transient voltage excitation control, and pure reactance network parameters.

Second, through the power-angle characteristic calculation and simulation, the maximum power output limit of the generator was obtained. It was verified that when the mechanical power input exceeds the peak value of the power-angle curve, the generator will lose the balance of acceleration and deceleration power, resulting in continuous rise of the power angle and out-of-synchronism instability, which directly reflects the physical mechanism of transient stability constraint.

In addition, the step disturbance test of mechanical power (10% increment at 1 s) shows that the system presents damped oscillation characteristics under small disturbance, and all state variables can converge to a new steady-state operating point stably. This proves that the system has good small-signal stability under the action of excitation control, and the damping and inertia parameters can effectively suppress electromechanical oscillation.

Finally, the simulation based on MATLAB and numerical solution method provides an intuitive and efficient way for power system dynamic analysis. The excitation control loop effectively maintains the generator transient voltage constant, which improves the steady-state operation performance and transient disturbance resistance of the system.

In summary, this experiment connects theoretical formulas with practical simulation, deepens the understanding of power system steady-state power flow, transient stability, small-signal stability and excitation control mechanism, and also improves the ability of modeling, programming and result analysis for power system dynamics.